Homework

Bioinformatics Engineering

ENBC455

Homework 12

Please complete the following problem. You must submit the following for each problem: a) the answer you are asked to return and b) Python code that can be run to verify your work.

1. Genome Assembly Using Reads

The goal of genome sequencing is to create contigs that are as long as possible. Thus, after fragment assembly, it is important to generate statistics quantifying how well-assembled our contigs are.

First and foremost, we demand a measure of what percentage of the assembled genome is made up of long contigs. Our first question is then: if we select contigs from our collection, how long do the contigs need to be to cover 50% of the genome?

PROBLEM

A directed cycle is simply a cycle in a directed graph in which the head of one edge is equal to the tail of the next (so that every edge in the cycle is traversed in the same direction).

For a set of DNA strings S and a positive integer k, let S_k denote the collection of all possible k-mers of the strings in S.

Given: A collection S of (error-free) reads of equal length (not exceeding 50 bp). In this dataset, for some positive integer k, the de Bruijn graph B_k on $S_{k+1} \cup S_{k+1}^{rc}$ consists of exactly two directed cycles.

Return: A cyclic superstring of minimal length containing every read or its reverse complement.

Sample Dataset

```
AATCT
TGTAA
GATTA
ACAGA
```

Sample Output

```
GATTACA
```

Hint: The reads "AATCT" and "TGTAA" are not present in the answer, but their reverse complements "AGATT" and "TTACA" are present in the circular string (GATTACA).